THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE ANNUAL EXAMINATION
041 BASIC MATHEMATICS

Time: 2 ½ Hours **November 2025**

Instructions

1. This paper consists of ten questions.
2. **Answer all questions**
3. Show clearly all the working and answers in the space provided.
4. Four figure mathematical tables, geometric instruments and graph papers may be used where necessary.
5. Any unauthorized materials are not allowed in the examination room.
6. Write your examination number on every page of your answer sheets.

QUESTION #	MARKS	INITIALS OF EXAMINER
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
Total		

1. (a) i. (a) Evaluate the following: $\frac{7}{8} \div \left(1\frac{1}{2} \times 1\frac{5}{9}\right)$.

(b) Use a number line to find the value of the following: $(-6) + 2$.

2. (a)Write the following numbers into decimal notation:

i. 2.555555555…

ii. 0.21212121…

iii. 1.15151515…

iv. 5.0000300003…

v. $\frac{4}{7}$

(b)Find the product of the answers in 2(a) (i) and (ii) in form of $\frac{a}{b}$, $where\ b \neq 0$

(c) Write:

i. 34.996 correct to 2 decimal places.

ii. 35.0483 correct to 3 significant figures.

3. (a) The following data shows the number of people who travelled by bus in seven days in Seypalm city as follows: 20 192, 18 800, 23 468, 25 477, 18 991, 260 060 and 22 645. Find the total number of passengers correct the answer t 2 significant figures.

(b) Represent $-5 < y \leq 2$ on a number line, where y is a real number.

4. (a) Solve the following simultaneous equations by combined method: $\begin{cases} x + \frac{y}{2} = 5 \\ \frac{x}{2} - \frac{y}{3} = \frac{1}{6} \end{cases}$

(b) Find the difference between the largest share and the smallest share when 60 000 shillings is shared into the ratio of 5: 3: 2.

5. (a) Find the equation and $y - intercept$ of a line passing through the following pairs of points$(4,4)\ and(-3,2)$.

(b) Graph the solution of the following: $|x - 3| \leq 5$

6. (a) Express 200 centimetres to 5 metres as a lowest ratio.

(b) Use the following linear equation to fill in the following table of values

x	-3	-2	-1	0	1	2	3
$y = -x + 1$							

7. (a) i. Simplifies the following: $\frac{1}{q}(qz + 4qz)$

ii. Solve: $\frac{2}{x} + 3 = 4$

(b) The sum of two numbers is 86while their difference is 14. Find the numbers.

8. (a) Solve graphically the following linear simultaneous equations: $\begin{cases} 2x + y = 5 \\ x + y = 3 \end{cases}$

(b) Write all numbers which are both multiples of 2 and 7 between $0 - 50$

9. (a) Write the place value of digit 2 from the product of 206 and 206.

(b)Describe three branches of mathematics.

10. (a)Subtract 5% of 300 000 shillings from 35% of 600 000 shillings.

(b) If point $(-6, n) is \ on \ the \ line \ 3x - 4y - 6 = 0$, find the value of n.